PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. The deflection in moving coil galvanometer falls from 25 divisions to 5 division when a shunt of 24 Ω is applied. The resistance of galvanometer coil will be :

(1) 100 Ω (2) 12 Ω

(3) 96 Ω (4) 48 Ω

Answer (3)

Sol. $i \propto$ number of divisions

$\therefore \quad 5(R_G) = (25 - 5)\,(24)$

$R_G = 96\ \Omega$

32. Match List-I with List-II

	List-I		List-II
a.	$\oint \vec{B}\cdot\vec{dl} = \mu_0 i_c + \mu_0\varepsilon_0 \frac{d\phi_E}{dt}$	(i)	Gauss' law for electricity
b.	$\oint \vec{E}\cdot\vec{dl} = \frac{d\phi_B}{dt}$	(ii)	Gauss' law for magnetism
c.	$\oint \vec{E}\cdot\vec{dA} = \frac{Q}{\varepsilon_0}$	(iii)	Faraday law
d.	$\oint B\cdot\vec{dA} = 0$	(iv)	Ampere - Maxwell las

Choose the correct answer from the options given below :

(1) a(iv), b(iii), c(i), d(ii) (2) a(ii), b(iii), c(i), d(iv)

(3) a(iv), b(i), c(iii), d(ii) (4) a(i), b(ii), c(iii), d(iv)

Answer (1)

Sol. $\oint \vec{E}\cdot\vec{dx} = \frac{Q}{\varepsilon_0}$ (Gauss's law)

Similarly,

$\oint \vec{E}\cdot\vec{dl} = \frac{d\phi_B}{dt}$ (Faraday law)

$\therefore$ a(iv), b(iii), c(i), d(ii)

33. At what distance above and below the surface of the earth a body will have same weight. (take radius of earth as *R*.)

(1) $\frac{\sqrt{5}R - R}{2}$ (2) $\sqrt{5}R - R$

(3) $\frac{R}{2}$ (4) $\frac{\sqrt{3}R - R}{2}$

Answer (1)

Sol. $g\frac{R^2}{(R+h)^2} = g\left(1 - \frac{h}{R}\right)$

$h = \left(\frac{\sqrt{5}-1}{2}\right)R$

34. A biconvex lens of refractive index 1.5 has a focal length of 20 cm in air. Its focal length when immersed in a liquid of refractive index 1.6 will be :

(1) +16 cm (2) +160 cm

(3) –160 cm (4) –16 cm

Answer (3)

Sol. $\frac{1}{f_{air}} = (1.5-1)\left(\frac{2}{R}\right) \Rightarrow R = 20$ cm

$\frac{1}{f_{liq}} = \left(\frac{1.5}{1.6} - 1\right)\left(\frac{2}{20}\right)$

$f_{liq} = -160$ cm



35. Two vessels *A* and *B* are of the same size and are at same temperature. *A* contains 1 g of hydrogen and *B* contains 1 g of oxygen. P_A and P_B are the pressure of the gases in *A* and *B* respectively, then $\frac{P_A}{P_B}$ is :

(1) 4 (2) 32

(3) 8 (4) 16

Answer (4)

Sol. $\because \quad PV = nRT$

$$n_A = \frac{1}{2}, \; n_B = \frac{1}{32}$$

$$\therefore \quad \frac{P_A}{P_B} = \frac{32}{2} = 16$$

36. The explosive in a hydrogen bomb is a mixture of ${}_1H^2$, ${}_1H^3$ and ${}_3Li^6$ in some condensed form. The chain reaction is given by

$${}_3Li^6 + {}_0n^1 \longrightarrow {}_2He^4 + {}_1H^3$$

$${}_1H^2 + {}_1H^3 \longrightarrow {}_2He^4 + {}_0n^1$$

During the explosion the energy released is approximately

[Given : M(Li) = 6.01690 amu, M($_1H^2$) = 2.01471 amu, M($_2He^4$) = 4.00388 amu, and 1 amu = 931.5 MeV]

(1) 22.22 MeV (2) 16.48 MeV

(3) 28.12 MeV (4) 12.64 MeV

Answer (1)

Sol. Q value = energy released

$$= c^2\left[\begin{matrix} m\left({}_3Li^6\right) + m\left({}_0n^1\right) - m\left({}_2He^4\right) - m\left({}_1H^3\right) \\ + m\left({}_1H^2\right) + m\left({}_1H^3\right) - m\left({}_2He^4\right) - m\left({}_0n^1\right) \end{matrix}\right]$$

= 22.22 MeV

37. If the radius of curvature of the path of two particles of same mass are in the ratio 3 : 4, then in order to have constant centripetal force, their velocities will be in the ratio of :

(1) $\sqrt{3} : 1$ (2) $2 : \sqrt{3}$

(3) $\sqrt{3} : 2$ (4) $1 : \sqrt{3}$

Answer (3)

Sol. $\frac{mv_1^2}{R_1} = \frac{mv_2^2}{R_2}$

$$\Rightarrow \quad \frac{v_1}{v_2} = \sqrt{\frac{R_1}{R_2}} = \frac{\sqrt{3}}{2}$$

38. A block of mass 100 kg slides over a distance of 10 m on a horizontal surface. If the co-efficient of friction between the surface is 0.4, then the work done against friction (in *J*) is :

(1) 4200 (2) 4000

(3) 4500 (4) 3900

Answer (2)

Sol. $W_f = |\mu mg\, x|$

$= 0.4 \times 100 \times 10 \times 10$

$= 4000$ J

39. The de-Broglie wavelength of an electron is the same as that of a photon. If velocity of electron is 25% of the velocity of light, then the ratio of K.E. of electron and K.E. of photon will be

(1) $\frac{1}{8}$ (2) $\frac{1}{1}$

(3) $\frac{1}{4}$ (4) $\frac{8}{1}$

Answer (1)



Sol. $\frac{v_e}{v_p} = \frac{1}{4}$ and $p_e = p_p$

$$\therefore \quad \frac{KE_e}{KE_p} = \frac{\frac{1}{2}(p_e)v_e}{(p_p)v_p} = \frac{1}{8}$$

40. A galvanometer having coil resistance 10Ω shows a full scale deflection for a current of 3*mA*. For it to measure a current of 8*A*, the value of the shunt should be :

(1) $3 \times 10^{-3}\Omega$ (2) $4.85 \times 10^{-3}\Omega$

(3) $3.75 \times 10^{-3}\Omega$ (4) $2.75 \times 10^{-3}\Omega$

Answer (3)

Sol. $i_G\,(G) = (i - i_G)\,s$

$$\therefore \quad s = 3.75 \times 10^{-3}\ \Omega$$

41. The resistance $R = \frac{V}{I}$ where $V = (200 \pm 5)$ *V* and *I* = (20 ± 0.2) A, the percentage error in the measurement of *R* is

(1) 3.5% (2) 3%

(3) 5.5% (4) 7%

Answer (1)

Sol. $\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$

$$\therefore \quad \frac{\Delta R}{R} \times 100 = 3.5\%$$

42. Two charges of 5*Q* and –2*Q* are situated at the points (3*a*, 0) and (–5*a*, 0) respectively. The electric flux through a sphere of radius '4*a*' having center at origin is

(1) $\frac{2Q}{\varepsilon_0}$ (2) $\frac{3Q}{\varepsilon_0}$

(3) $\frac{7Q}{\varepsilon_0}$ (4) $\frac{5Q}{\varepsilon_0}$

Answer (4)

Sol. $\phi = \frac{q_{enclosed}}{\varepsilon_0}$

$$= \frac{5Q}{\varepsilon_0}$$

43. Given below are two statements:

Statement I: If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in hot water.

Statement II: If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in cold water.

In the light of the above statements, choose the most appropriate from the options given below

(1) Statement **I** is false but Statement **II** is true

(2) Both Statement **I** and Statement **II** are true

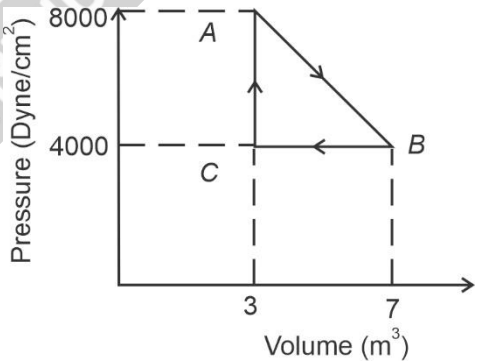
(3) Both Statement **I** and Statement **II** are false

(4) Statement **I** is true but Statement **II** is false

Answer (4)

Sol. As temperature increases, surface tension decreases. And [$h \propto$ surface tension]

44. A thermodynamic system is taken from an original state *A* to an intermediate state *B* by a linear process as shown in the figure. It's volume is then reduced to the original value from *B* to *C* by an isobaric process. The total work done by the gas from *A* to *B* and *B* to *C* would be:



(1) 1200 J (2) 2200 J

(3) 600 J (4) 33800 J

Answer (None)



Sol. Work done $= \frac{1}{2}(4)(4000)\times 10^{-1}$ J

$= 800$ J

No option is matching.

45. A convex mirror of radius of curvature 30 cm forms an image that is half the size of the object. The object distance is

(1) 15 cm (2) 45 cm

(3) –45 cm (4) –15 cm

Answer (4)

Sol. $m = \frac{1}{2} = \frac{-v}{u}$

$\Rightarrow \quad v = -\frac{u}{2}$

Now

$\frac{1}{v} + \frac{1}{u} = \frac{1}{15}$

$\frac{-2}{u} + \frac{1}{u} = \frac{1}{15}$

$\boxed{u = -15 \text{ cm}}$

46. A body starts moving from rest with constant acceleration covers displacement S_1 in first $(p-1)$ seconds and S_2 in first p seconds. The displacement $S_1 + S_2$ will be made in time:

(1) $(2p^2 - 2p + 1)s$

(2) $(2p + 1)s$

(3) $\sqrt{(2p^2 - 2p + 1)s}$

(4) $(2p - 1)s$

Answer (3)

Sol. $S_1 = \frac{1}{2}a(p-1)^2$

$S_2 = \frac{1}{2}a(p)^2$

$\therefore \quad S_1 + S_2 = \frac{1}{2}a(p^2 + 1 - 2p + p^2)$

$= \frac{1}{2}at^2$

$\therefore \quad t = \sqrt{2p^2 - 2p + 1}$

47. A capacitor of capacitance 100 μF is charged to a potential of 12 V and connected to a 6.4 mH inductor to produce oscillations. The maximum current in the circuit would be:

(1) 2.0 A (2) 1.2 A

(3) 1.5 A (4) 3.2 A

Answer (3)

Sol. $i = Q\omega$

$= CV\sqrt{\frac{1}{LC}}$

$= \sqrt{\frac{C}{L}}V$

$= \sqrt{\frac{10^{-4}}{64\times 10^{-4}}}(12)$

$= 1.5$ A

48. The potential energy function (in J) of a particle in a region of space is given as $U = (2x^2 + 3y^2 + 2z)$. Hare x, y and z are in meter. The magnitude of x-component of force (in N) acting on the particle at point $P(1, 2, 3)m$ is:

(1) 6 (2) 8

(3) 2 (4) 4

Answer (4)



Sol. $F_x = \frac{-\partial u}{-\partial x} = -4x$

$\therefore$ magnitude = 4 N

49. The electric current through a wire varies with time as $I = I_0 + \beta t$, where I_0 = 20 A and β = 3 A/s. The amount of electric charge crossed through a section of the wire in 20 s is:

(1) 800 C (2) 1600 C

(3) 80 C (4) 1000 C

Answer (4)

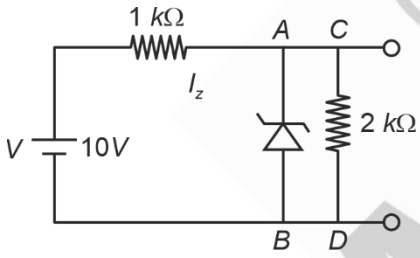
Sol. $\Delta_q = \int i\,dt$

$= \int_0^{20} (20+3t)\,dt$

$= \left[20t + \frac{3t^2}{2}\right]_0^{20}$

= 1000 C

50. In the given circuit, the breakdown voltage of the Zener diode is 3.0 V. What is the value of I_z?



(1) 7 mA (2) 5.5 mA

(3) 3.3 mA (4) 10 mA

Answer (2)

Sol.



$i_z = \left(\frac{7V}{1\,k\Omega} - \frac{3V}{2\,k\Omega}\right)$

= 5.5 mA

SECTION - B

Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

51. In a test experiment on a model aeroplane in wind tunnel, the flow speeds on the upper and lower surfaces of the wings are 70 ms^{-1} and 65 ms^{-1} respectively. If the area is 2 m^2, the lift of the wing is _____N.

(Given density of air = 1.2 kg m^{-3})

Answer (810)

Sol. $F = \Delta P(A)$

$= \frac{1}{2}\rho(v_u^2 - v_L^2)(A)$

$= \frac{1}{2}\times 1.2(70^2 - 65^2)2$

= 810 N

52. A cylinder is rolling down on an inclined plane of inclination 60°. It's acceleration during rolling down will be $\frac{x}{\sqrt{3}}$ m/s^2, where x = _____ (use g = 10 m/s^2)

Answer (10)

Sol. $a = \frac{g\sin\theta}{1+\frac{I}{mR^2}}$ $\left(\because I = \frac{mR^2}{2}\right)$

$= \frac{2g\sin 60°}{3}$

$= \frac{10}{\sqrt{3}}$ m/s^2



53. A ball rolls off the top of a stairway with horizontal velocity u. The steps are 0.1 m high and 0.1 m wide. The minimum velocity u with which that ball just hits the step 5 of the stairway will be $\sqrt{x}$ ms^{-1} where x = _____ [use g = 10 m/s^2].

Answer (2)

Sol. To hit 5th step it should just cross the 4th step.

$\therefore$ Range = 0.4 m, Height = 0.4 m.

$$\therefore R = u\sqrt{\frac{2H}{g}}$$

$$0.4 = u\sqrt{\frac{2(0.4)}{10}}$$

$$u = \sqrt{2} \text{ m/s.}$$

54. The magnetic potential due to a magnetic dipole at a point on its axis situated at a distance of 20 cm from its center is 1.5 × 10^{-5} T m. The magnetic moment of the dipole is _____ A m^2. (Given: $\frac{\mu_0}{4\pi}$ = 10^{-7} T m A^{-1})

Answer (6)

Sol. $V_{axial} = \frac{\mu_0}{4\pi}\frac{M}{r^2}$

$$\therefore 1.5\times10^{-5} = 10^{-7}\frac{M}{(20\times10^{-2})^2}$$

$\Rightarrow M$ = 6 A m^2

55. A electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet S having surface charge density +σ. The electron at t = 0 is at a distance of 1 m from S and has a speed of 1 m/s. The maximum value of σ if the electron strikes S at t = 1 s is $\alpha\left[\frac{m\in_0}{e}\right]\frac{C}{m^2}$, the value of α is _____.

Answer (8)

Sol. e^- should be thrown away from sheet.

$$\because s = ut + \frac{1}{2}at^2$$

$$-1 = 1(1) + \frac{1}{2}\left(\frac{-\sigma e}{2\in_0 m}\right)(1)^2$$

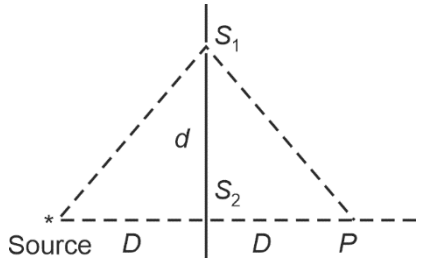
$$\therefore \sigma = \frac{8\in_0 m}{e}$$

56. When the displacement of a simple harmonic oscillator is one third of its amplitude, the ratio of total energy to the kinetic energy is $\frac{x}{8}$, where x = _____.

Answer (9)

Sol. $$\frac{\text{T.E}}{\text{K.E}} = \frac{\frac{1}{2}KA^2}{\frac{1}{2}K(A^2-x^2)} = \frac{9}{8}$$

57. In a double slit experiment shown in figure, when light of wavelength 400 nm is used, dark fringe is observed at P. If D = 0.2 m, the minimum distance between the slits S_1 and S_2 is _____ mm.



Answer (0.2)



Sol. $\Delta x = \frac{\lambda}{2}$ (for min. d)

$$2\left[\sqrt{D^2+d^2}-D\right]=200\times10^{-9}$$

$$D\left(1+\frac{d^2}{2D^2}\right)-D=10^{-7}$$

$$\frac{d^2}{20}=10^{-7}$$

$$\Rightarrow\ d^2=2\times(0.2)\times10^{-7}=4\times10^{-8}\ \text{m}$$

$$d=2\times10^{-4}\ \text{m} = 0.2\ \text{mm}$$

58. A 16 Ω wire is bend to form a square loop. A 9 V battery with internal resistance 1 Ω is connected across one of its sides. If a 4 μF capacitor is connected across one of its diagonals, the energy stored by the capacitor will be $\frac{x}{2}$ μJ. where x = _____.

Answer (81)

Sol. $i_{\text{battery}}=\frac{9}{3+1}=\frac{9}{4}$ A

$$(\Delta V)_{\text{cap.}}=\frac{9}{4}\left(\frac{1}{4}\right)(8)=\frac{9}{2}\ \text{V}$$

$$\Delta U=\frac{1}{2}CV^2$$

$$=\frac{1}{2}(4)\left(\frac{9}{2}\right)^2\ \mu\text{J}=\frac{81}{2}\ \mu\text{J}$$

$x = 81$

59. When a hydrogen atom going from $n = 2$ to $n = 1$ emits a photon, its recoil speed is $\frac{x}{5}$ m/s. Where x = _____. (Use mass of hydrogen atom = 1.6 × 10^{-27} kg)

Answer (17)

Sol. $\Delta E=13.6\left(1-\frac{1}{4}\right)=\frac{3(13.6)}{4}$ eV

now, $v=\frac{p}{m}=\frac{E}{mC}$

$$=\frac{\frac{3}{4}(13.6)\times1.6\times10^{-19}}{3\times10^{8}\times1.6\times10^{-27}}=\frac{17}{5}\ \text{m/s}$$

60. A square loop of side 10 cm and resistance 0.7 Ω is placed vertically in east-west plane. A uniform magnetic field of 0.20 T is set up across the plane in north east direction. The magnetic field is decreased to zero in 1 s at a steady rate. Then magnitude of induced emf is $\sqrt{x}\times10^{-3}$ V. The value of x is _____.

Answer (2)

Sol. $\varepsilon_{\text{ind}}=\frac{\Delta\phi}{\Delta t}=\frac{BA\cos45°-0}{\Delta t}$

$$=\frac{0.2(10^{-2})\left(\frac{1}{\sqrt{2}}\right)}{1}$$

$$=\sqrt{2}\times10^{-3}\ \text{V}$$

$\boxed{x=2}$



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. Match List I with List II.

	List I (Substances)		List II (Element Present)
A.	Ziegler catalyst	I.	Rhodium
B.	Blood pigment	II.	Cobalt
C.	Wilkinson catalyst	III.	Iron
D.	Vitamin B_{12}	IV.	Titanium

Choose the correct answer from the options given below.

(1) A-II, B-IV, C-I, D-III (2) A-III, B-II, C-IV, D-I

(3) A-II, B-III, C-IV, D-I (4) A-IV, B-III, C-I, D-II

Answer (4)

Sol. Ziegler catalyst – $TiCl_4 \cdot Al(C_2H_5)_3$

Blood pigment – Compound containing iron

Wilkinson catalyst – $[RhCl(PPh_3)_3]$

Vitamin B_{12} – Compound containing cobalt

$\therefore$ Correct match is A-IV, B-III, C-I, D-II.

62. Given below are two statements :

Statement I : The electronegativity of group 14-elements from Si to Pb, gradually decreases.

Statement II : Group 14 contains non-metallic, metallic, as well as metalloid elements.

In the light of the above statements, choose the **most appropriate** from the options given below.

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

Answer (4)

Sol. Statement-I is false because electronegativity of Group-14 elements from Si to Pb is almost constant, *i.e.*, 1.8.

Statement-II is correct because Group-14 contains non-metals, metalloids as well as metals.

Carbon	–	Non-metal
Silicon	–	Metalloid
Germanium	–	Metalloid
Tin	–	Metal
Lead	–	Metal

63. In chromyl chloride test for confirmation of Cl^- ion, a yellow solution is obtained. Acidification of the solution and addition of amyl alcohol and 10% H_2O_2 turns organic layer blue indicating formation of chromium pentoxide. The oxidation state of chromium in that is

(1) +6 (2) +10

(3) +5 (4) +3

Answer (1)

Sol. The reactions involved in chromyl chloride test and subsequent formation of chromium pentaoxide are as follows :

$$Cr_2O_7^{2-} + 6H^+ + 4Cl^- \longrightarrow \underset{\text{(Red)}}{2CrO_2Cl_2(g)} + 3H_2O$$

$$CrO_2Cl_2(g) + 4NaOH(aq) \longrightarrow \underset{\text{(Yellow)}}{Na_2CrO_4(aq)} + 2NaCl + 2H_2O$$

$$\underset{\text{(Yellow)}}{2Na_2CrO_4(aq)} + 2H^+ \longrightarrow \underset{\text{(Orange)}}{Na_2Cr_2O_7(aq)} + 2Na^+ + H_2O$$

$$Na_2Cr_2O_7 + 4H_2O_2 + 2H^+ \longrightarrow \underset{\substack{\text{(Blue in}\\ \text{amyl alcohol)}}}{2CrO_5} + 5H_2O + 2Na^+$$

Oxidation state of chromium in CrO_5 is +6.

